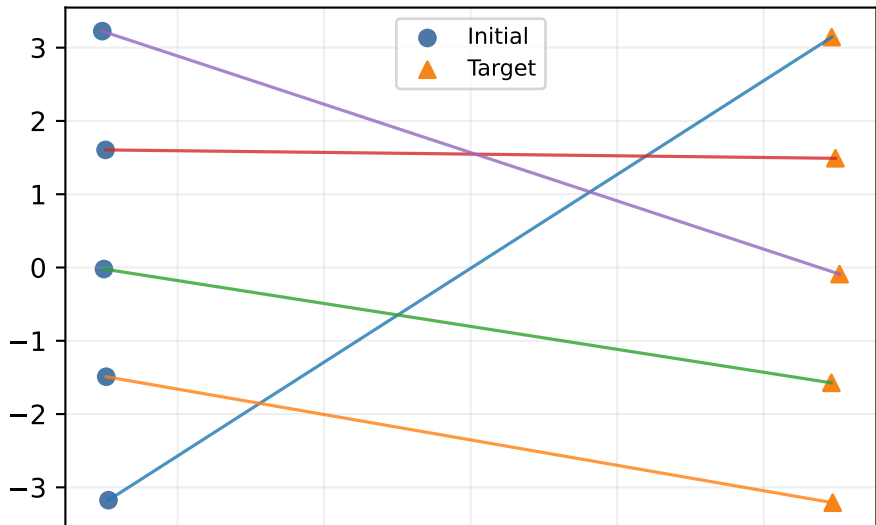
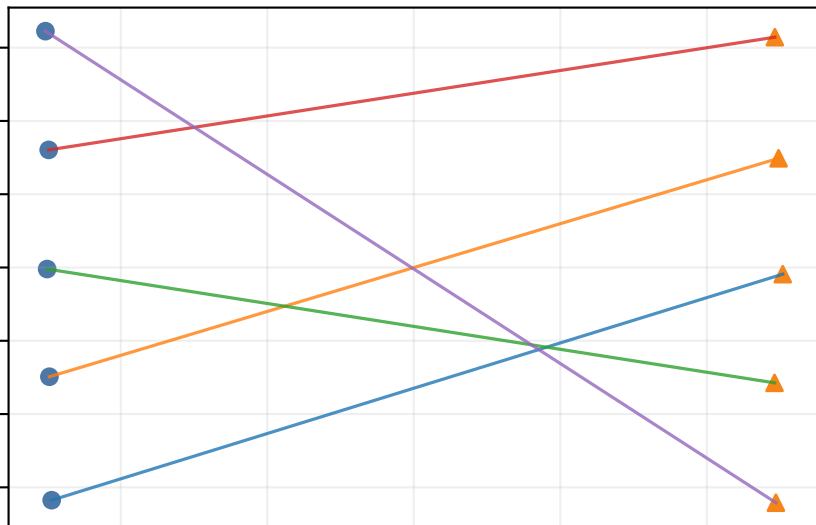


Crossing-prone Trial 5: Assignment Paths (XY Projection)

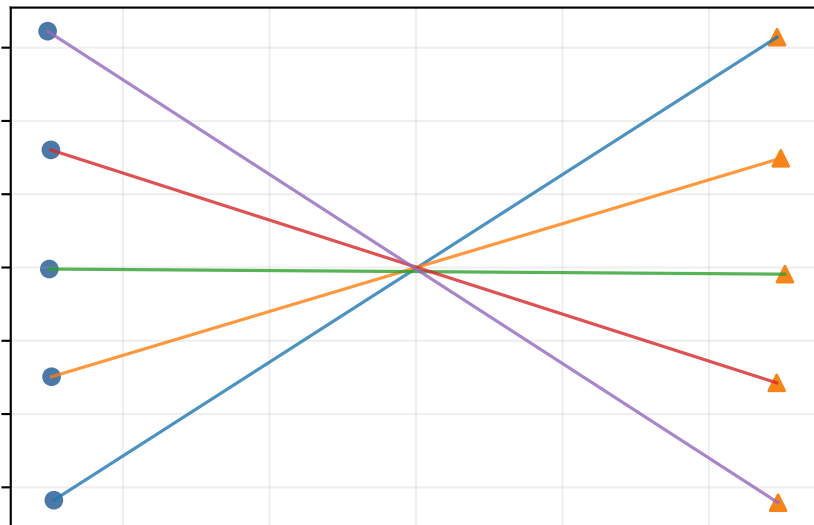
Random  
XY=5, violations=175



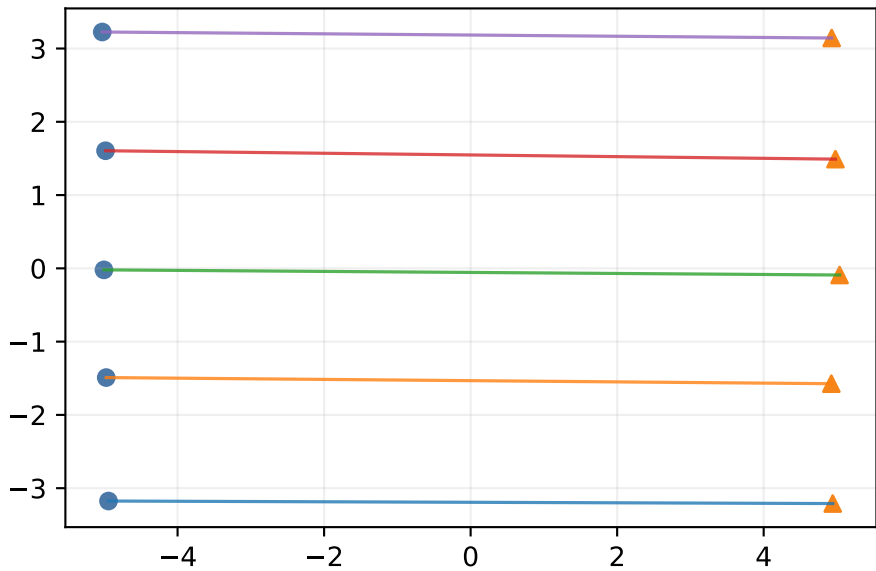
Nearest Neighbor  
XY=6, violations=96



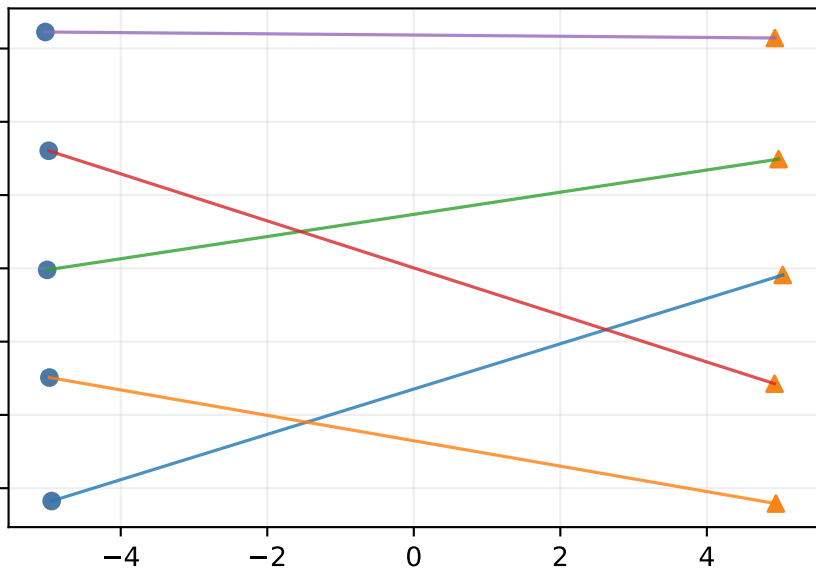
Hungarian-Distance  
XY=10, violations=301



Hungarian + crossing penalty  
XY=0, violations=300



Hungarian + safety-aware local swap  
XY=3, violations=0



X (m)